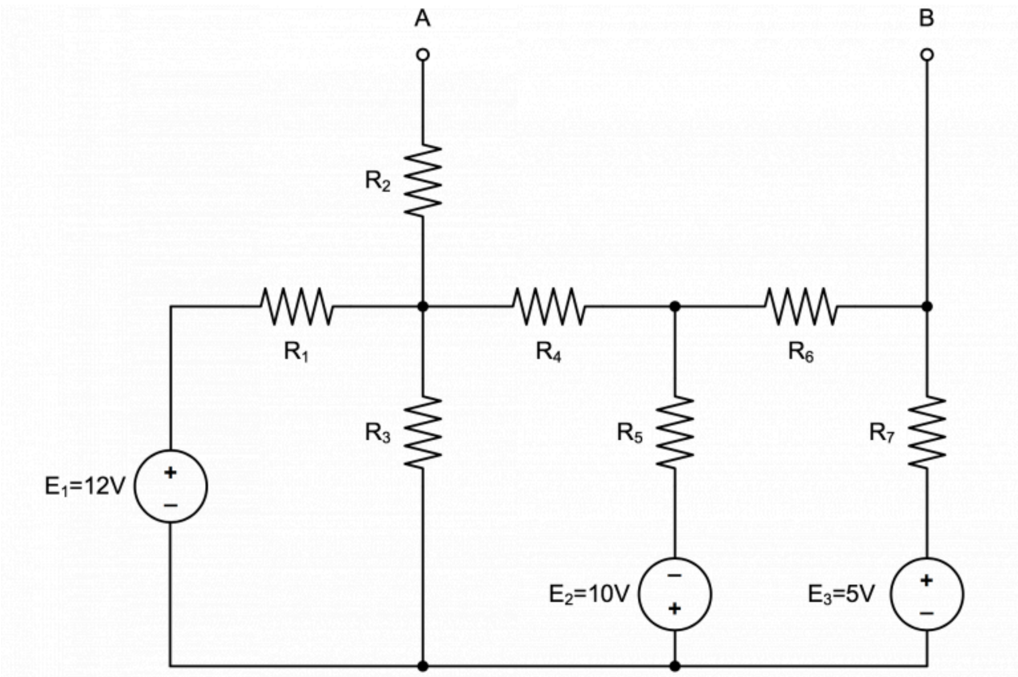
1 Lab 2 Calculations

1.A Nodal & Mesh Analysis, Thevenin's Theorem and Superposition

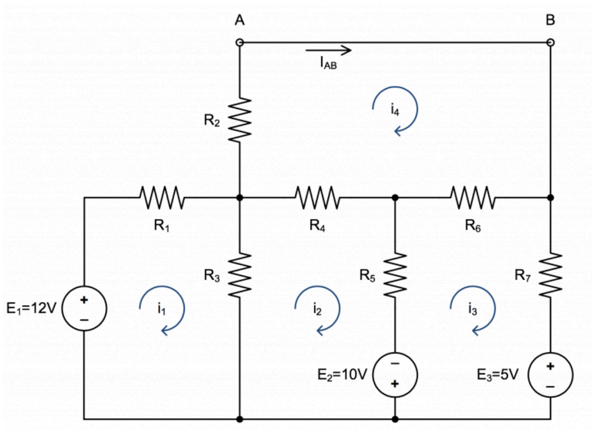
For the circuit in Figure 3-1, all 7 resistors have different resistance values. For each resistor, choose any resistance value within the range of $1k\Omega$ to $56k\Omega$ unless specified otherwise.

$R_1 = 1k\Omega,$
$R_2 = 2k\Omega,$
$R_3 = 3k\Omega,$
$R_4 = 10k\Omega,$
$R_5 = 11k\Omega,$
$R_6 = 12k\Omega,$
$R_7 = 13k\Omega$



1.A.1

Short AB, as shown in Figure 3-2(a). Use mesh analysis to calculate the voltage across each resistor and the current through AB, I_{AB}.



(a)

KVL@i_1

$$12V = R_1 i_1 + R_3(i_1 - i_2)$$

KVL@i_2

$$10V = R_3(i_2 - i_1) + R_4(i_2 - i_4) + R_5(i_2 - i_3)$$

KVL@i_3

$$-5V - 10V = R_5(i_3 - i_2) + R_6(i_3 - i_4) + R_7 i_3$$

KVL@i_4

$$0 = R_4(i_4 - i_2) + R_2 i_4 + R_6(i_4 - i_3)$$

MATLAB code:

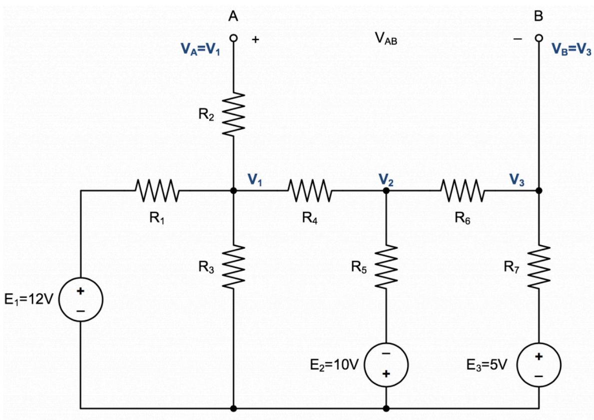
```
% Given values
R1 = 1e3; % Ohms
R2 = 2e3; % Ohms
R3 = 3e3; % Ohms
R4 = 10e3; % Ohms
R5 = 11e3; % Ohms
R6 = 12e3; % Ohms
R7 = 13e3; % Ohms

syms i1 i2 i3 i4;
% KVL equations
eq1 = 12 == R1*i1 + R3*(i1 - i2);
eq2 = 10 == R3*(i2 - i1) + R4*(i2 - i4) + R5*(i2 - i3);
eq3 = -5 - 10 == R5*(i3 - i2) + R6*(i3 - i4) + R7*i3;
eq4 = 0 == R4*(i4 - i2) + R2*i4 + R6*(i4 - i3);
% Solve for currents
currents = solve([eq1, eq2, eq3, eq4], [i1, i2, i3, i4])
```

$$i_1 = 3.912mA$$
$$i_2 = 1.215mA$$
$$i_3 = 148\mu A$$
$$i_4 = 581\mu A$$
$$i_{\mathrm{AB}} = i_4 = 581\mu A$$

1.A.2

Leave AB open, as shown in Figure 3-2(b). Use nodal analysis to calculate the voltage across each resistor as well as the voltage across AB, V_{AB}.



(b)

KCL@V_1

$$0 = \frac{12V - V_1}{R_1} + \frac{0 - V_1}{R_3} + \frac{V_2 - V_1}{R_4}$$

KCL@V_2

$$0 = \frac{V_1 - V_2}{R_4} + \frac{-10 - V_2}{R_5} + \frac{V_3 - V_2}{R_6}$$

KCL@V_3

$$0 = \frac{V_2 - V_3}{R_6} + \frac{-5 - V_3}{R_7}$$

MATLAB Code:

```
% Given values
R1 = 1e3; % Ohms
R2 = 2e3; % Ohms
R3 = 3e3; % Ohms
R4 = 10e3; % Ohms
R5 = 11e3; % Ohms
R6 = 12e3; % Ohms
R7 = 13e3; % Ohms

syms V1 V2 V3;
% KCL equations
eq1 = 0 == (12 - V1)/R1 + (0 - V1)/R3 + (V2 - V1)/R4;
eq2 = 0 == (V1 - V2)/R4 + (-10 - V2)/R5 + (V3 - V2)/R6;
eq3 = 0 == (V2 - V3)/R6 + (-5 - V3)/R7;
% Solve for node voltages
voltages = solve([eq1, eq2, eq3], [V1, V2, V3])
```

$$\begin{aligned} V_1 &= 8.287V \\ V_2 &= -1.214V \\ V_3 &= -3.031V \\ V_{\mathrm{AB}} &= V_1 - V_3 = 11.318V \end{aligned}$$

1.A.3

Find Thevenin's and Norton's Equivalent using the results from the previous two steps.

$$\begin{aligned} V_{\mathrm{TH}} &= V_{\mathrm{AB}} = 11.318V \\ R_{\mathrm{TH}} &= \frac{V_{\mathrm{AB}}}{I_{\mathrm{AB}}} = 19.47k\Omega \\ I_{\mathrm{N}} &= I_{\mathrm{AB}} = 581\mu A \\ R_{\mathrm{N}} &= R_{\mathrm{TH}} = 19.47k\Omega \end{aligned}$$

1.A.4

Connect a resistor between nodes A and B. Use Thevenin's theorem to calculate the current through this resistor for the following 3 resistor values: $1k\Omega$, $2.2k\Omega$ and $4.7k\Omega$.

$$\begin{aligned} I_1 &= \frac{V_{\mathrm{TH}}}{R_{\mathrm{TH}} + 1k\Omega} = 553\mu A \\ I_2 &= \frac{V_{\mathrm{TH}}}{R_{\mathrm{TH}} + 2.2k\Omega} = 522\mu A \\ I_3 &= \frac{V_{\mathrm{TH}}}{R_{\mathrm{TH}} + 4.7k\Omega} = 468\mu A \end{aligned}$$

1.A.5

Leave AB open. Find the voltage across AB, V_{AB} for the following 3 cases:

Case 1: E_1 is turned on while both E_2 and E_3 are turned off.

```
eq1 = 0 == (12 - V1)/R1 + (0 - V1)/R3 + (V2 - V1)/R4;
eq2 = 0 == (V1 - V2)/R4 + (0 - V2)/R5 + (V3 - V2)/R6;
eq3 = 0 == (V2 - V3)/R6 + (0 - V3)/R7;
```

$$\begin{aligned} V_1 &= 8.633V \\ V_2 &= 3.739V \\ V_3 &= 1.944V \\ V_{\mathrm{AB}} &= V_1 - V_3 = 6.689V \end{aligned}$$

Case 2: E_2 is turned on while both E_1 and E_3 are turned off.

```
eq1 = 0 == (0 - V1)/R1 + (0 - V1)/R3 + (V2 - V1)/R4;
eq2 = 0 == (V1 - V2)/R4 + (-10 - V2)/R5 + (V3 - V2)/R6;
eq3 = 0 == (V2 - V3)/R6 + (0 - V3)/R7;
```

$$\begin{aligned} V_1 &= -283mV \\ V_2 &= -4.060V \\ V_3 &= -2.111V \\ V_{\mathrm{AB}} &= V_1 - V_3 = 1.828V \end{aligned}$$

Case 3: E_3 is turned on while both E_1 and E_2 are turned off.

```
eq1 = 0 == (0 - V1)/R1 + (0 - V1)/R3 + (V2 - V1)/R4;
eq2 = 0 == (V1 - V2)/R4 + (0 - V2)/R5 + (V3 - V2)/R6;
eq3 = 0 == (V2 - V3)/R6 + (-5 - V3)/R7;
```

$$\begin{aligned} V_1 &= -62mV \\ V_2 &= -893mV \\ V_3 &= -2.864V \\ V_{\mathrm{AB}} &= V_1 - V_3 = 2.802V \end{aligned}$$

Superposition:

Adding the three voltages demonstrates voltage superposition. The actual voltage can be calculated by summing the individual contributions from each voltage source. The $1mV$ discrepancy is due to accumulated rounding errors in the calculations.

$$\Sigma V_{\mathrm{AB}} = 6.689V + 1.828V + 2.802V = 11.319V \approx 11.318V$$